

ANSWER KEY

EECS 489 Final Exam

Fall 2024

Name: _____

Uniqname: _____

INSTRUCTIONS

Welcome to the EECS 489 Fall 2024 Final.

- This exam is 120 minutes long and consists of both multiple choice and free response questions. Partial credit will be awarded where applicable, so please show all work and reasoning in your responses.
- This is a closed-book exam. You may use one note sheet, 8.5"×11", double-sided, with your name on it.
- Mark all solutions in the space provided. Any answers given outside the boxes will not be graded.
- Clearly fill your multiple choice answers. Erase completely if you change your answer. There is exactly one answer for every question.
- Bubble in your answer on the bar **above** the question. If your circled and bubbled answers are different, we will choose the answer that is bubbled in.
- There are questions on both sides of each page.
- The total points on the exam add up to 88.

Good luck!

When you are finished with the exam, sign below acknowledging the Honor Code

"I have neither given nor received unauthorized aid on this examination, nor have I concealed any violations of the Honor Code."

ANSWER KEY

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ANSWER KEY

Multiple Choice

(2 points each)

☐ ☒ B ☐ C ☐ D

1. Suppose a link has a bandwidth of 20 Mbps. There are 4 flows with demands of 2 Mbps, 4 Mbps, 10 Mbps, and 6 Mbps over this link. What is the allocated bandwidth for each of the flows (in Mbps) using Min-Max fairness? Allocated bandwidth is listed in the same order as demands.

- a) **2, 4, 8, 6**
- b) 2, 4, 9, 5
- c) 5, 5, 5, 5
- d) $\frac{20}{11}, \frac{40}{11}, \frac{100}{11}, \frac{60}{11}$

Explanation: As there are 4 flows with a total bandwidth of 20 Mbps, the initial fair share for each flow is $\frac{20 \text{ Mbps}}{4} = 5 \text{ Mbps}$.

The first two flows have demands less than the fair share, so the remainder of their allocation – $(5 \text{ Mbps} - 2 \text{ Mbps}) + (5 \text{ Mbps} - 4 \text{ Mbps}) = 4 \text{ Mbps}$ – is distributed equally among the two remaining flows. Thus, the fair share for the two remaining flows increases to $5 \text{ Mbps} + \frac{4 \text{ Mbps}}{2} = 7 \text{ Mbps}$.

Now, the fourth flow's demand can be fully satisfied, and the remaining $7 \text{ Mbps} - 6 \text{ Mbps} = 1 \text{ Mbps}$ is allocated to the third flow, which now has an allocation of $7 \text{ Mbps} + 1 \text{ Mbps} = 8 \text{ Mbps}$. Thus, the correct answer is **a) 2, 4, 8, 6**.

☐ A ☒ B ☐ C ☐ D

2. An IPv4 packet with a total size of 3000 bytes (including the IP header) is being sent from a device through two LANs. The first LAN has an MTU of 1500 bytes. What is the minimum number of fragments that are required to send this packet? You may assume that all IP headers are 20 bytes.

- a) 2
- b) **3**
- c) 4
- d) 5

Explanation: The packet has $3000 - 20 = 2980$ bytes of data. An IP packet in the first LAN can have at most $1500 - 20 = 1480$ bytes of data. Thus, we require fragmentation into a minimum of $\lceil \frac{2980}{1480} \rceil = 3$ packets.

ANSWER KEY

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3. Suppose a router uses a **longest prefix match** algorithm for routing. If its routing table has the following entries:

Prefix	Netmask	Next Hop
192.168.0.0	255.255.0.0 (/16)	Router A
192.168.10.0	255.255.255.0 (/24)	Router B
192.168.10.128	255.255.255.128 (/25)	Router C
192.168.10.127	255.255.255.192 (/26)	Router D

Which next hop will an IP packet with a destination IP 192.168.10.130 go to?

- a) Router A
- b) Router B
- c) **Router C**
- d) Router D

Explanation: A longest prefix match algorithm checks whether (prefix & netmask) == (dest & netmask) for each row in the routing table, returning the row with the longest netmask. In this case, this is Router C.

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4. Which one of these is **not** an example of a mechanism that can be described as router-assisted congestion control?

- a) Explicit Congestion Notification (ECN)
- b) Weighted Fair Queuing
- c) Rate Control Protocol (RCP)
- d) **Additive Increase, Multiplicative Decrease (AIMD)**

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5. Which of the following [IPv4 header field]: [Purpose] pairings is **incorrect**?

- a) Time-to-live: Prevent routing loops
- b) **Identification: Detect Corruption**
- c) Fragment Offset: Handle Large Packets
- d) Protocol: Identify Packet Type

ANSWER KEY

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6. Which of the following statements regarding IPv6 is **incorrect**?
- a) IPv6 reduces the need for NAT (Network Address Translation)
 - b) **IPv6 uses the “Flow Label” field to enable fragmentation**
 - c) IPv6 lacks checksum functionality
 - d) IPv6 has fixed-length headers

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7. When implementing the WTP-Base protocol in Project 3, Efe accidentally introduces a single bug. However, when he sanity-checks his code by transferring a large file, he manages to do so correctly. Which one of the following bugs is consistent with this behavior?

Recall that WTP-Base uses cumulative ACKs, with the receiver buffering out-of-order packets that are within the window. For each option, assume that no other bugs exist. Assume the file is large enough that it requires at least twice the `window_size` number of packets to transmit. Assume the link has non-zero propagation delay, transmission delay, and packet loss rate.

- a) `wReceiver` sends selective ACKs, sending an ACK with sequence number i in response to a packet with sequence number i .
- b) When `wReceiver` calls `recvfrom` using a UDP socket to get a new packet, it supplies a buffer of size 1 in order to read byte-by-byte.
- c) `wSender` numbers packets starting at 1 instead of 0.
- d) **`wSender` advances the sender-side window only when an ACK arrives that indicates the last packet in the window was received.**

Explanation: (A) will cause the sender-side window to never move. (B) will cause incomplete data to be received; data exceeding the buffer size will be dropped, as UDP is a message-based protocol rather than a stream-based protocol. (C) will cause the receiver to keep asking for Packet 0, never advancing the receiver-side window. (D) will work, although it might lead to degraded performance.

ANSWER KEY

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8. Aditya is now implementing the WTP-OPT protocol for Project 3; he also accidentally introduces a single bug. He notices that his code successfully manages to transfer files sometimes, but not always. Which one of the following bugs is consistent with this behavior?

Recall that WTP-OPT uses selective ACKs. For each option, assume that no other bugs exist. Assume the file requires at least twice the `window_size` number of packets to transmit. Assume the link has non-zero propagation delay, transmission delay, and packet loss rate.

- a) A `sleep` call in `wReceiver` causes ACKs to be sent after 1 second of receiving the packet, despite the sender-side timeout being 500ms.
- b) `wReceiver` sends cumulative ACKs, sending an ACK with sequence number equal to the next expected sequence number.
- c) `wSender` resends all packets in the window instead of just the unacknowledged packets when any packet times out.
- d) **`wSender` shifts the window by one packet every time an ACK is received, ignoring the sequence number of the ACK.**

Explanation: (A) would not cause sporadic behavior. The file would always be transmitted, as `wSender` is equipped to handle arbitrary delays in ACKs.

(B) would not cause sporadic behavior for the opposite reason – the file transfer would never work, as the sender would never receive acknowledgement for the first packet (with sequence number 0).

(C) would not cause sporadic behavior; the receiver is equipped to handle duplicate packets.

(D) would cause sporadic behavior; it would only work if no data packets are ever dropped, and would not work otherwise.

ANSWER KEY

● (B) (C) (D)

9. Which of the following is an accurate statement about software-defined networking (SDN)?

- a) **SDN involves a centralized control plane and distributed data plane.**
- b) An SDN controller routes incoming packets to different routers for load-balancing.
- c) Routers using SDN are slower than traditional routers as a remote SDN controller must be consulted to route every packet rather than a local routing table.
- d) SDN is only applicable in data centers and enterprise networks.

Explanation:

(A) is correct; the control plane is centralized as a SDN controller centrally plans routing flows for individual routers. On the other hand, individual routers still route incoming packets on their own, creating a distributed data plane.

(B) is incorrect; this describes the function of a load-balancer, which is not a function of an SDN controller.

(C) is incorrect; once a routing table is installed on a router, it makes data plane decisions independently without consulting the SDN controller, with no difference in performance compared to a traditional router.

(D) is incorrect; although the most frequent use cases of SDN are in places where centralized control is easy (like data centers and enterprise networks), SDN principles can be applied to IoT networks, edge devices, etc.

● (B) (C) (D)

10. Which of the following correctly describes a difference between *infrastructure mode* and *ad hoc mode* in a wireless network?

- a) **Infrastructure mode requires a central access point, while ad hoc mode allows devices to communicate directly without a central access point.**
- b) Infrastructure mode creates point-to-point communication, while ad hoc mode enables broadcast communication.
- c) Ad hoc mode is more scalable than infrastructure mode as there is less overhead created from infrastructure concerns.
- d) Wireless access points in infrastructure mode are connected by wire to L2 switches; on the other hand, wireless access points in ad hoc mode are connected wirelessly to L3 routers.

ANSWER KEY

☐ A ☐ B ☐ C ☒

11. Which of the following is not an application of tunneling?
- a) Creating a Virtual Private Network (VPN)
 - b) Supporting new protocols, such as IPv6
 - c) Optimizing network traffic by forcing packets to use specific paths
 - d) **Transmitting large packets that do not fit in a single Ethernet frame**

☐ A ☒ B ☐ C ☐ D

12. Which of the following is **not** a disadvantage of Network Address Translation (NAT)?
- a) NAT adds additional complexity for machines behind different NAT devices to establish direct connections, as each device no longer has a unique IP address.
 - b) **NAT reduces security, as packets can no longer be encrypted in order for the NAT device to translate IP addresses.**
 - c) NAT adds some translation overhead, potentially increasing latency and consuming processing power.
 - d) NAT can complicate network troubleshooting by hiding IP addresses.

☐ A ☐ B ☐ C ☒

13. What is a disadvantage of using topology-aware addressing in data center networks?
- a) It reduces network scalability as the addressing scheme grows too complex.
 - b) It complicates routing decisions and increases overhead by necessitating complex forwarding tables that align with network topology.
 - c) It leads to inefficient use of available bandwidth and may cause congestion in certain parts of the network.
 - d) **It requires regular updates to the addressing scheme to reflect the dynamic changes in the network topology.**

ANSWER KEY

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14. Suppose that you are a router in a network that uses distance vector updates for calculating the shortest path to a destination with poisoned reverse. Suppose that your currently calculated distance vector to other routers is

$$\begin{pmatrix} k & u & v & z \\ 5 & 7 & 2 & 10 \end{pmatrix}$$

Router k then sends you the following updated distance vector message:

$$\begin{pmatrix} k & u & v & z & m \\ 0 & 3 & 5 & 4 & 7 \end{pmatrix}$$

Which of the following is the correctly updated distance vector that belongs to you, including any routers that were previously unknown to us?

a) $\begin{pmatrix} k & u & v & z & m \\ 5 & 7 & 2 & 10 & 7 \end{pmatrix}$

b) $\begin{pmatrix} k & u & v & z & m \\ 5 & 7 & 2 & 9 & 12 \end{pmatrix}$

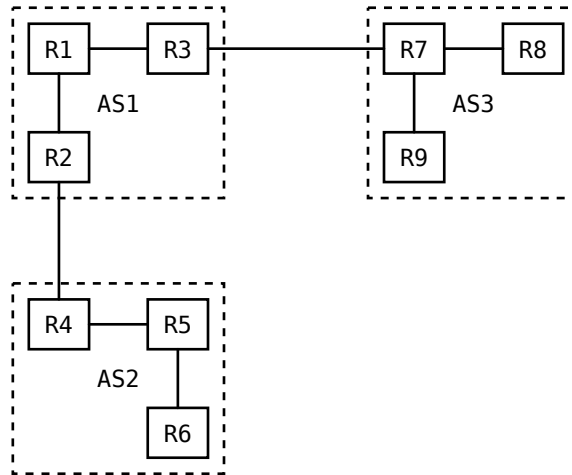
c) $\begin{pmatrix} k & u & v & z & m \\ 5 & 3 & 2 & 10 & 7 \end{pmatrix}$

d) $\begin{pmatrix} k & u & v & z & m \\ 5 & 8 & 10 & 9 & 12 \end{pmatrix}$

ANSWER KEY

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15. For this question, refer to the below network diagram. Each dashed square represents an AS. The ASes and the routers within use BGP to communicate about network paths. Assume that all ASes in the diagram have peer-peer relationships and use standard Gao-Rexford rules to decide which paths to communicate.



How do R1, R4, R6, and R8 learn about a subnet accessible only through R3?

Option	R1	R4	R6	R8
I	IGP	eBGP	eBGP	iBGP
II	iBGP	eBGP	iBGP	iBGP
III	iBGP	eBGP	IGP	IGP
IV	IGP	eBGP	iBGP	iBGP

- a) I
- b) II
- c) III
- d) **IV**

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16. Which of the following is **not** an accurate statement regarding Intra-AS routing?

- a) Link-state routing tends to have smaller messages than distance-vector routing.
- b) Distance-vector routing can take much longer to converge than link-state routing.
- c) Distance-vector routing is more similar to BGP routing than link-state routing.
- d) **The poisoned reverse heuristic can help both distance-vector and link-state routing eliminate count-to-infinity scenarios.**

ANSWER KEY

☐ A ☐ B ☒ C ☐ D

17. Which of the following is **not** an accurate statement regarding IP and MAC addresses?

- a) MAC addresses are associated with individual network interfaces rather than devices.
- b) There are more IPv6 addresses than MAC addresses.
- c) **IPv4 addresses are always permanently assigned, but can easily be spoofed.**
- d) An ARP request is intended to discover the MAC address corresponding to an IP address.

☒ A ☐ B ☐ C ☐ D

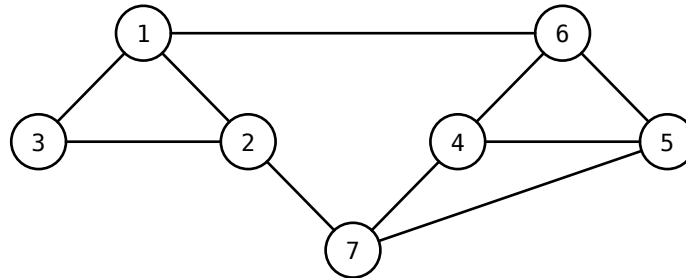
18. Which of the following is the Dynamic Host Configuration Protocol (DHCP) used for?

- a) **Discovering the subnet mask of the current network**
- b) Discovering the IP address that maps to a certain URL
- c) Sending error messages when packets are unable to be delivered
- d) Creating a spanning tree to prevent broadcast storms on a switched Ethernet network

ANSWER KEY

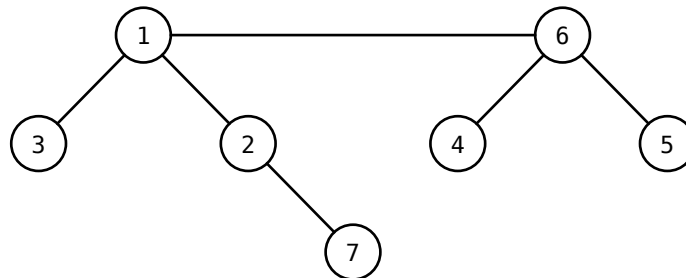
● B C D

19. Consider the following graph of nodes in a switched Ethernet network. Suppose these nodes construct a spanning tree amongst themselves using the algorithm learned in class, electing the node with the smallest identifier as the root. Which of the following edges will **not** be part of the spanning tree?



- a) (4, 7)
- b) (5, 6)
- c) (1, 3)
- d) (4, 6)

Explanation:



We do not need to trace through the entire algorithm; the idea here is to only preserve the edges that are on the shortest path from any node to Node 1. This graph has the special property that every node has a unique shortest path to Node 1, so there is no ambiguity in doing so.

ANSWER KEY

Short Answer

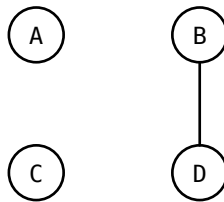
20. Poisoned Reverse

(10 points)

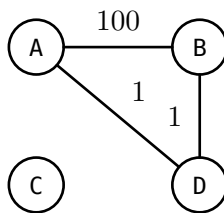
20.1 (4 points) Suppose a distance-vector protocol is being used for intra-AS routing. Please draw any additional edges between the nodes as you see fit and assign costs to all edges such that deleting the edge BD creates

- a count-to-infinity scenario between A and D if poisoned reverse is not used.
- no issue if poisoned reverse is used.

Briefly explain your answer.



Answer:



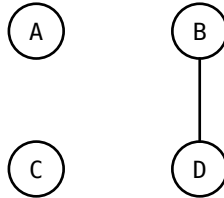
Initially, A has the distance vector $\begin{pmatrix} b & d \\ 2 & 1 \end{pmatrix}$ while D has the distance vector $\begin{pmatrix} a & b \\ 1 & 1 \end{pmatrix}$. When the edge BD is deleted, D may decide to route to B through A instead, updating its distance vector to $\begin{pmatrix} a & b \\ 1 & 3 \end{pmatrix}$. This is a problem because A routes to B through D, so a routing loop is created. A then updates its distance vector to $\begin{pmatrix} b & d \\ 4 & 1 \end{pmatrix}$, and then D updates its distance vector based on this update, and so on until the AB link is utilized (which takes a while).

If poisoned reverse is used, A's distance vector is advertised to D as $\begin{pmatrix} b & d \\ \infty & \infty \end{pmatrix}$. So, when the BD link is deleted, D updates its distance vector and advertises it to A as $\begin{pmatrix} a & b \\ \infty & \infty \end{pmatrix}$, which in turn causes A to route directly through the AB link instead. A's distance vector is now advertised to D as $\begin{pmatrix} b & d \\ 100 & \infty \end{pmatrix}$, which allows D to update its distance vector to $\begin{pmatrix} a & b \\ 1 & 101 \end{pmatrix}$.

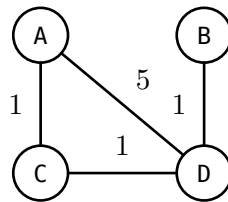
ANSWER KEY

20.2 (4 points) Suppose a distance-vector protocol is being used **with** poisoned reverse. Please draw any additional edges between the nodes as you see fit and assign costs to all edges such that deleting the edge BD still creates a count-to-infinity scenario between A and D .

Briefly explain your answer.



Answer:



When the BD edge is deleted, D attempts to route to B through A . In turn, A routes to B through C and then D . This creates a routing loop between A , C , and D . As D is not the next-hop for the A to B route, A does not advertise its cost to B as infinite to D , creating this problem.

20.3 (2 points) Does BGP routing encounter count-to-infinity scenarios? Why or why not? Explain in 2-3 sentences.

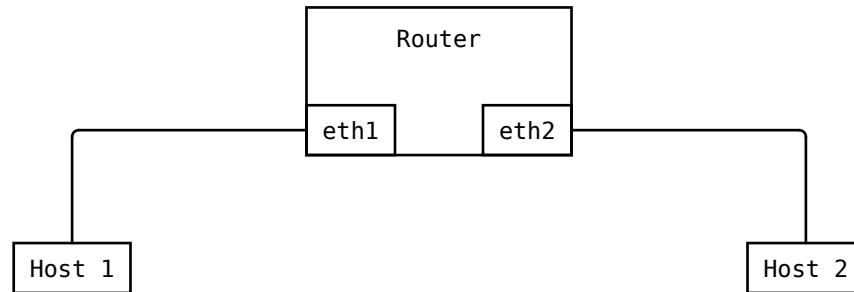
Answer: Nodes in BGP use the **path-vector** protocol, advertising a path to another node rather than a distance. This prevents a count-to-infinity scenario as routing loops never occur, as nodes can use the path information from their neighbors to avoid creating these loops.

ANSWER KEY

21. Sending a Packet

(15 points)

Consider the following network topology:



Host 1 and Host 2 are on different local area networks (LANs), with the router connecting the two hosts. The router and the hosts each have their own ARP Cache to keep track of IP-to-MAC mappings; all ARP Caches are initially empty. This is very similar to the setup in Project 4.

Some information about the network interfaces being used are given below:

Interface	MAC Address	IP Address
Host 1	A	B
Host 2	C	D
eth 1	E	F
eth 2	G	H

ANSWER KEY

21.1 (5 points) Host 1 would like to send a UDP packet to Host 2. Please indicate all packets that will arrive and leave the router in chronological order in the following table. You may not need to use all rows.

In/Out of Router	Packet Type (ARP Request/ARP Reply/ICMP/UDP)	Router Interface (eth1/eth2)
In	ARP Request	eth1
Out	ARP Reply	eth1
In	UDP	eth1
Out	ARP Request	eth2
In	ARP Reply	eth2
Out	UDP	eth2

21.2 (5 points) Fill in the blank fields for the indicated layers on the UDP packet that Host 1 would send to communicate with Host 2 in 21.1.

Ethernet Layer	
Source MAC	A
Destination MAC	E

IP Layer	
Source IP	B
Destination IP	D

ANSWER KEY

21.3 (5 points) Now, Host 1 decides to send a ping (echo) request to Host 2. Once Host 2 receives this ping request, it will reply. Please indicate all packets that will enter and leave the router as a result of this exchange (including the reply) in chronological order.

Assume that any ARP entries cached from the previous exchange are still valid, and no other ARP entries have been created. As in Project 4, assume that replying to echo requests does not require IP-to-MAC resolution. Once again, you may not need to use all rows.

In/Out of Router	Packet Type (ARP Request/ARP Reply/ICMP/UDP)	Router Interface (eth1 / eth2)
In	ICMP	eth1
Out	ICMP	eth2
In	ICMP	eth2
Out	ARP Request	eth1
In	ARP Reply	eth1
Out	ICMP	eth1

ANSWER KEY

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ANSWER KEY

22. “Switching” Up P4

In Project 4, we implemented a router that learned the MAC addresses of different routers using ARP requests. In lecture, we discussed how bridges on local networks instead use self-learning to know how to route incoming packets on a local network. In this question, you will be implementing a self-learning bridge.

Refer to the reference sheet on the following page for the structure of the class.

You will be implementing the `LearningBridge::handlePacket` function. You may assume that all other functions have been implemented and function correctly. For simplicity, assume that links are connected in a spanning-tree configuration, so you do not need to worry about broadcast storms.

Answer:

```
void LearningBridge::handlePacket(EthernetPacket packet, std::string port) {
    associatePortWithMac(packet.source_mac, port);

    auto outPort = getPortForMac(packet.dest);
    if (outPort) {
        sendPacket(packet, *outPort);
        return;
    }

    const auto& allPorts = getAllPorts();
    for (const auto& outPort : allPorts) {
        if (outPort != port) {
            sendPacket(packet, outPort);
        }
    }
}
```

ANSWER KEY

Reference for Question 22

```
struct EthernetPacket {
    mac_addr source_mac;
    mac_addr dest_mac;
    std::vector<uint8_t> data;
};

/**
 * @class LearningBridge A switch that learns the ports different hosts are located on
 * by inspecting incoming packets.
 */
class LearningBridge {
public:
    /**
     * @brief Called when an ethernet packet is received on a port. This function ensures
     * that the switch "learns" from the incoming packet if possible.
     *
     * If the destination is not associated with a port, the packet should be broadcasted.
     * If the destination is associated with a port, the packet should only be sent out
     * on that port.
     */
    void handlePacket(EthernetPacket packet, std::string port);
private:
    /**
     * @brief Sends a packet out on the specified port
     */
    void sendPacket(EthernetPacket packet, std::string port);
    /**
     * @brief Associates a host with the MAC address specified with the specified port
     */
    void associatePortWithMac(mac_addr mac, std::string port);
    /**
     * @brief Return the port associated with the host with the given MAC address,
     * if such an association exists
     */
    std::optional<std::string> getPortForMac(mac_addr mac);

    /**
     * @brief Return a vector of all the ports that exist on the switch
     */
    const std::vector<std::string>& getAllPorts();

    // Member data structures not shown
};
```

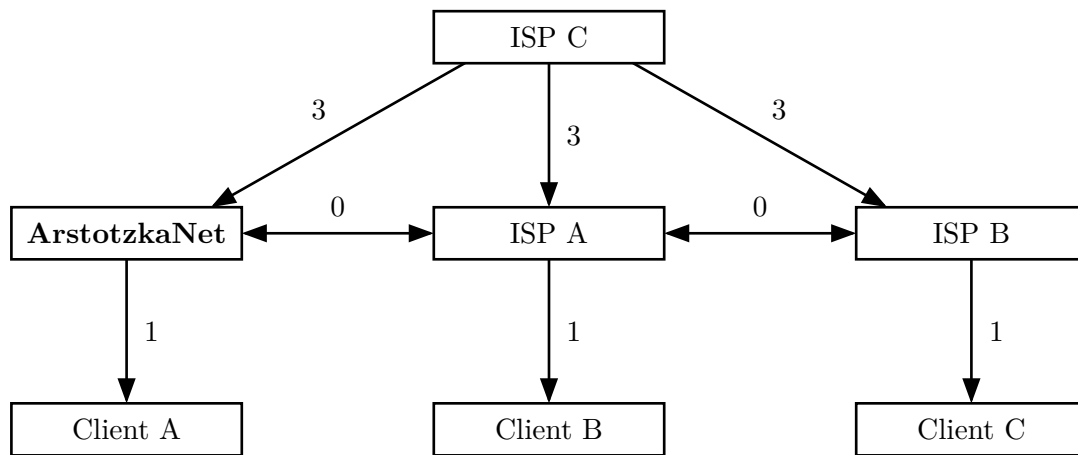
ANSWER KEY

23. BGP, Please

(10 points)

Congratulations! You have been promoted to a position where you are in charge of BGP configurations for a global internet service provider (ISP) called ArstotzkaNet. You are responsible for configuring and enforcing BGP agreements such that all of the ISP's routers only route packets through autonomous systems (ASes) that they are allow to use for routing. You should assume that Gao-Rexford rules are being followed.

The BGP graph for your ISP is given in the following diagram. The edge weights represent the cost of routing packets through each connection.



$A \leftrightarrow B$: A and B have a peering connection
 $A \rightarrow B$: A is a provider of B

23.1 (3 points) Name the cheapest path that Client A can use to communicate with Client B. If there is a cheaper path that Client A cannot use because of BGP rules, write down the path and the reason it cannot be used.

Answer: The shortest path is Client A $\rightarrow$ ArstotzkaNet $\rightarrow$ ISP A $\rightarrow$ Client B

23.2 (3 points) Name the cheapest path that Client A can use to communicate with Client C. If there is a cheaper path that Client A cannot use because of BGP rules, write down the path and the reason it cannot be used.

Answer: The shortest path is Client A $\rightarrow$ ArstotzkaNet $\rightarrow$ ISP C $\rightarrow$ ISP B $\rightarrow$ Client C.

There exists a shorter path Client A $\rightarrow$ ArstotzkaNet $\rightarrow$ ISP A $\rightarrow$ ISP B $\rightarrow$ Client C, but the Gao-Rexford rules disallow peering connections from being used by non-peers.

ANSWER KEY

23.3 (4 points) Suppose that the government in your country has ordered that access to IP addresses within the `umich.edu` AS be blocked. Assume the `umich.edu` AS is a customer of ISP C that is not shown in this diagram.

Explain how ArstotzkaNet can take advantage of BGP to cause both Client A and ISP A to not be able to access the `umich.edu` AS (at least temporarily). For your answer to this question, ArstotzkaNet need not follow Gao-Rexford rules.

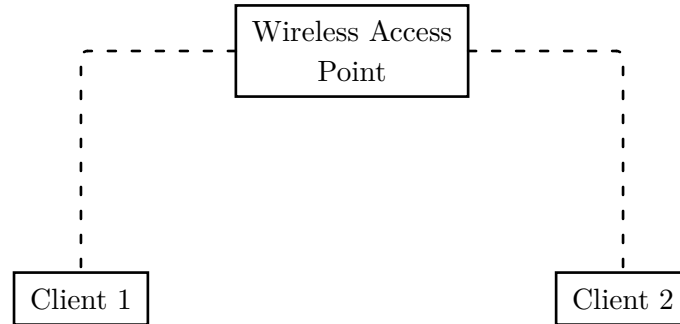
Answer: ArstotzkaNet can advertise to all of its customers and peers (i.e. Client A and ISP A) that there exists a shorter path to `umich.edu` ASes through itself than the actual shortest path. Because BGP is simply a protocol that allows sharing paths between ASes and not a protocol that provides any security or verification of such routes, other ASes will simply believe this path and attempt to route packets through ArstotzkaNet, who is free to simply drop them.

ANSWER KEY

24. Look Ma, No Wires!

(15 points)

Consider the below 802.11 wireless network setup below.



Assume that:

- The bandwidth of each wireless link is 1 MB/s.
- The one-way propagation delay between either client and the wireless access point is 0.5 seconds.
- The clients are not in direct range of each other.
- A client has a timeout of 1 RTT (i.e. 1s) for backing off after sending a packet if the expected response is not received.
- No packets are dropped or lost.

Both Client 1 and Client 2 select backoff times for CSMA/CA according to a function $B(i, k)$ such that

$$B(i, k) = 2(k + 1) \cdot i$$

where i is the number of the client, and k is the number of times the client has backed off previously.

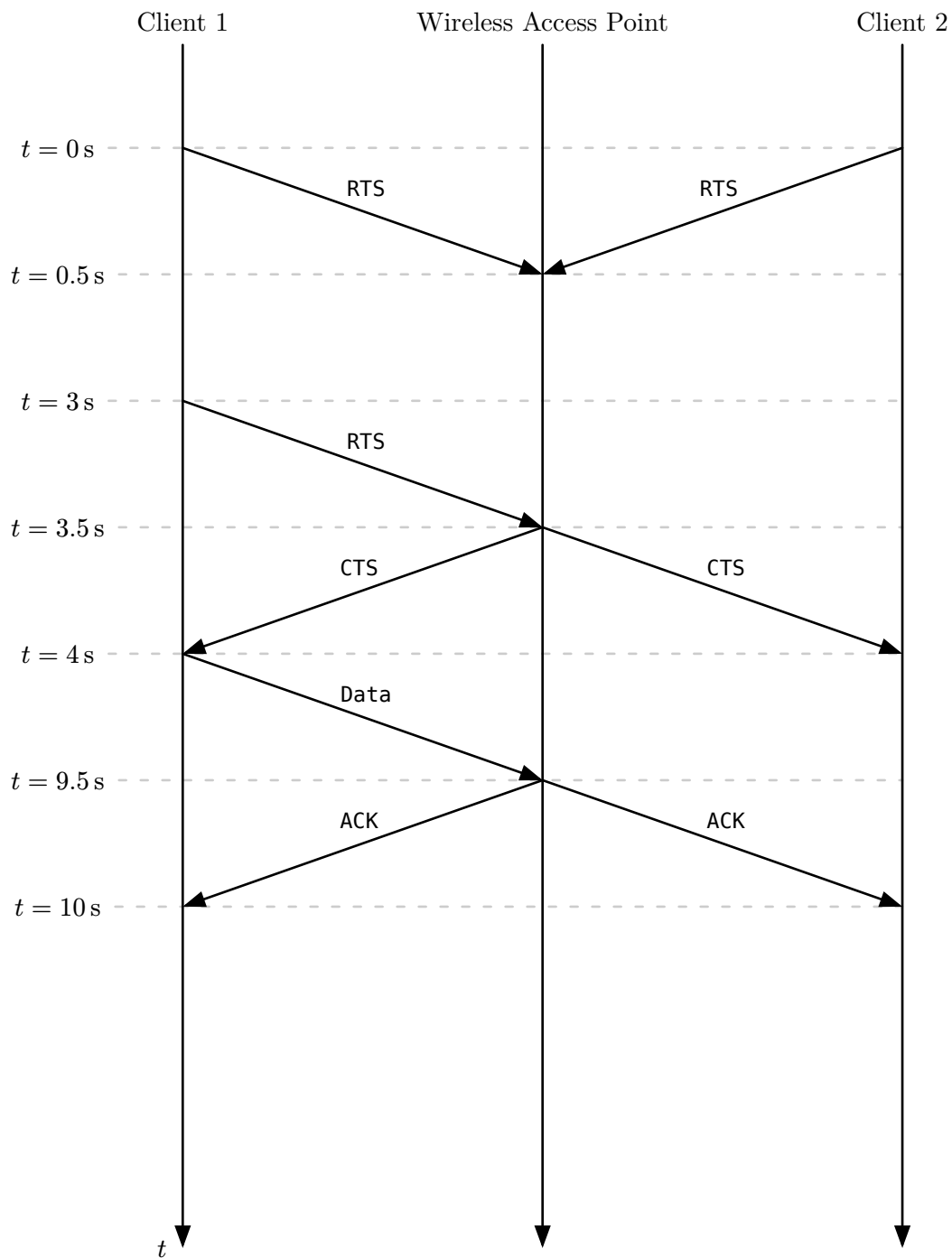
For example, if Client 1 backs off for the first time, $i = 1$ and $k = 0$.

ANSWER KEY

24.1 (9 points) Both Client 1 and Client 2 would both like to send 5 MB of data to the wireless access point. Both clients attempt to start transmitting at $t = 0$.

Below, draw the order in which different collision avoidance messages and actual data are sent until Client 1 completes its transmission, including receiving the acknowledgement message.

- As usual, time is on the vertical axis. The vertical axis does not need to be to-scale.
- Make sure to clearly indicate the type of message, the time transmission begins, the time the message is fully received, and the direction of the message.
- Keep in mind that wireless is a broadcast medium.

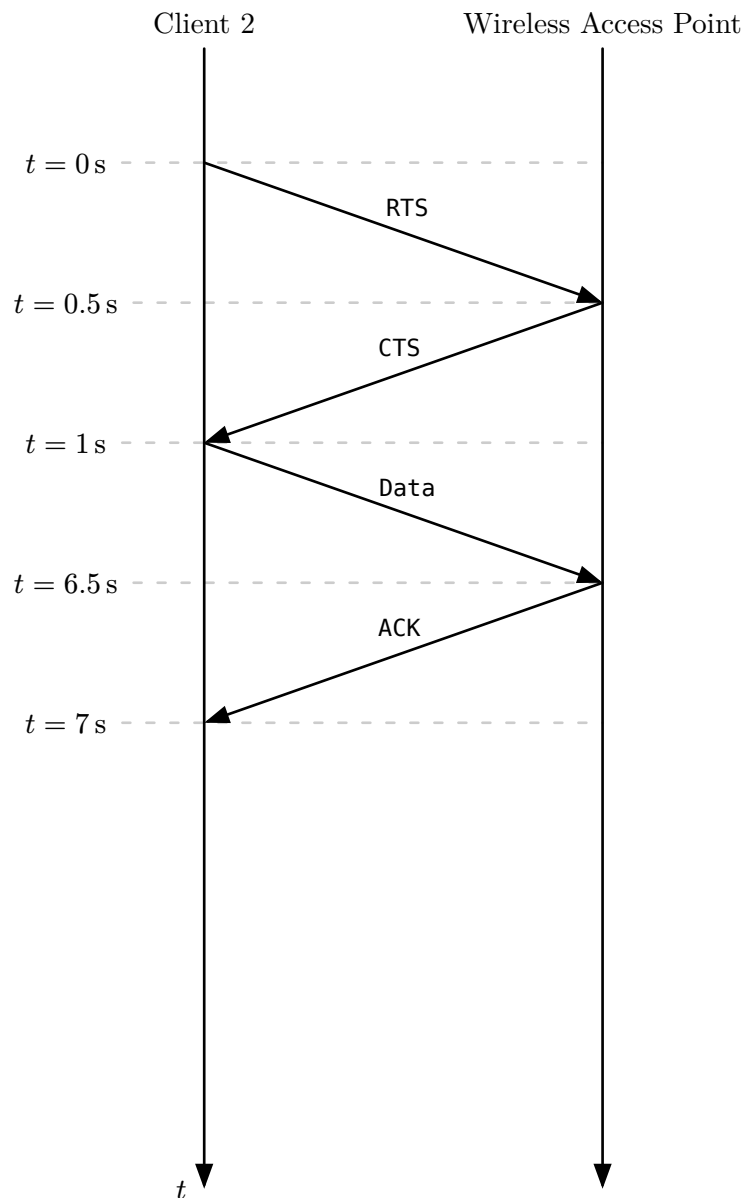


ANSWER KEY

24.2 (6 points) After Client 1 finishes its transmission, Client 2 attempts to transmit its 5 MB of data again.

Below, draw the order in which different collision avoidance messages and actual data are sent until Client 2 completes its transmission, including receiving its acknowledgement message.

- You should use $t = 0$ as the start time for Client 2's transmission (i.e. do **not** continue from the final time on question 24.1).
- As usual, time is on the vertical axis. The vertical axis does not need to be to-scale.
- Make sure to clearly indicate the type of message, the time transmission begins, the time the message is fully received, and the direction of the message.
- Keep in mind that wireless is a broadcast medium.



ANSWER KEY

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ANSWER KEY

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